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SENSOR ELEMENT, TEST DEVICE, AND METHOD FOR TESTING DATA CARRIERS HAVING A SPIN RESONANCE FEATURE

Abstract

A sensor element for testing a flat-surface data carrier having a spin resonance feature. The sensor element includes a magnetic core with an air gap, into which the flat-surface data carrier can be inserted for testing, a polarization device for generating a static magnetic flux in the air gap, a resonator device for exciting the spin resonance feature of the data carrier to be tested in the air gap, having at least one stripline resonator fed by a signal source, and a modulation device for generating a time-varying magnetic modulation field in the air gap parallel to the static magnetic field. The modulation device has a plurality of modulation coils, which are designed and configured for generating different modulation frequencies so that the modulated magnetic field generated by the modulation device, together with the polarization device, has different modulation frequencies at different locations within the air gap.

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Background/Summary

[0001] The invention relates to a sensor element for checking the authenticity of a flat-surface data carrier, in particular a banknote, having a spin resonance feature. The invention also relates to a test device having such a sensor element and to a method for testing authenticity using such a sensor element or such a test device.
[0002] Data carriers, such as value or identification documents, but also other valuable objects, such as brand-name articles, are often provided with security elements that allow the data carriers to be authenticated and that also serve as protection against unauthorized reproduction. It is well known in the field of machine authentication to use security elements with spin resonance features to secure documents and other data carriers. The security elements are provided with substances that have a spin resonance signature. The spin resonance signatures that can be used for authenticity testing include, in particular, nuclear magnetic resonance (NMR) effects, electron spin resonance (ESR) effects, and ferromagnetic resonance (FMR) effects.
[0003] In the process of checking banknotes, three different magnetic fields are usually generated in the measuring range of a banknote processing machine, for example, to detect the spin resonance signatures. This is specifically a quasi-static polarization field B.sub.0, which runs parallel to the axial direction (z direction) of the air gap of a magnetic circuit. A second magnetic field is formed by a modulation field B.sub.mod, which also runs parallel to the z-axis and typically has a frequency f mod in the kHz range. For excitation of transitions between the split spin energy levels of the spin resonance signature substances, an excitation field B.sub.1 is provided, which is polarized perpendicular to the B.sub.0 direction. The excitation field oscillates at the resonance frequency of the material, which is also referred to as the Larmor frequency, and which is proportional to the polarization field B.sub.0.
[0004] To generate the polarization field B.sub.0, a magnetic circuit is often used that directs the magnetic flux of permanent magnets and/or coils to an air gap in which the testing of the flat-surface data carriers takes place.
[0005] A high-frequency resonator, for example a stripline resonator, is used for generating the excitation field B.sub.1. A detector diode is used to measure the RF power reflected by the resonator. If a test specimen is in resonance at a coupled-in frequency, the resonator quality changes, and with it the power reflected by the resonator. Due to the field modulation B.sub.mod, the exact value of the Larmor frequency of the test specimen oscillates, and the measurement signal is amplitude modulated with f.sub.mod. The spectral power distribution at the input of the detector diode then shows, in addition to the central microwave carrier frequency, modulation peaks offset by ±f.sub.mod, which carry the desired spin resonance information.
[0006] When checking the authenticity of a data carrier, there is often a need to be able to operate multiple independent resonators with the same frequency, for example, in order to be able to detect the spin resonance information in a spatially resolved manner. Conventionally, each resonator requires an independent microwave circuit for the detection and evaluation of the measurement signal. These microwave circuits are typically realized on circuit boards by means of SMD

components and striplines. A large amount of installation space is required for multiple independent circuits. If this installation space is not available, the various circuits can quickly be affected by crosstalk causing signal distortion to occur.

[0007] In addition, the functionality of many elements in a microwave circuit is defined by the geometry of the elements involved. Especially at high frequencies, i.e. short wavelengths, the manufacturing tolerance has a strong effect on the functionality of the circuit. For example, if multiple identical microwave circuits are to be set up in parallel, for example for the above-mentioned spatially resolved measurement of spin resonances, the nominally identical microwave circuits can differ considerably in their functionality in practice and make reliable spatially resolved measurement more difficult.

[0008] Against this background, the object of the invention is to avoid the disadvantages of the prior art and in particular to provide a sensor element of the type mentioned above, which allows improved detection of the spin resonance feature of a flat-surface data carrier.

[0009] This object is achieved by means of the features of the independent claims. Developments of the invention are the subject of the dependent claims.

[0010] The invention provides a sensor element for testing, in particular testing the authenticity, of a flat-surface data carrier having a spin resonance feature. The flat-surface data carrier can be formed by a banknote, for example. The sensor element contains a magnetic core with an air gap, into which the flat-surface data carrier can be inserted for testing, a polarization device for generating a static magnetic flux in the air gap, and a resonator device for exciting the spin resonance feature of the data carrier to be tested in the air gap, having at least one stripline resonator fed by a signal source. The spin resonance feature is preferably an ESR feature.

[0011] The sensor element also contains a modulation device for generating a time-varying magnetic modulation field in the air gap parallel to the static magnetic field. The modulation device comprises a plurality of modulation coils, which are designed and configured for generating different modulation frequencies, so that the modulated magnetic field generated by the modulation device together with the polarization device has different modulation frequencies at different locations within the air gap.

[0012] In principle, stripline resonators are characterized in particular in that their sensitive region is very easily accessible and that they have a very high filling factor for flat samples, such as those formed by the banknotes to be tested. The stripline resonators are sometimes referred to below as resonators purely for brevity.

[0013] The resonator device is also designed in particular for detecting spin resonance signals of the spin resonance feature. The resonator device can in particular record a response signal of the spin resonance feature and output it to a detector. The spin resonances can be determined, for example, with a continuous wave (CW) method, a pulsed method, or a rapid scan method.

[0014] In an advantageous embodiment, it is provided that the modulation coils of the modulation device are arranged offset next to one another. Each modulation coil then generates a spatial region of the modulation field with its respective modulation frequency.

[0015] Advantageously, at least one or preferably even all modulation coils of the modulation device is/are formed by planar coils, which have one or more turns around an axial direction of the air gap in a plane. The said plane with the one or more turns is advantageously perpendicular to the axial direction (z direction) of the air gap, i.e. the direction between the pole surfaces of the magnetic core adjacent to the air gap. Thus, the modulation field is advantageously aligned parallel to the polarization field.

[0016] The modulation coils are advantageously all arranged in the same plane on a common coil carrier, in particular a common printed circuit board.

[0017] In an advantageous embodiment of the invention, the modulation coils are designed and configured for generating modulation frequencies which are not in a simple integer ratio to each other, in particular not in a ratio of 1:n, with n a natural number less than 6. In other words, the

generated modulation frequencies of each two modulation coils are not in a simple ratio of 1:2, 1:3, 1:4, 1:5, or a reciprocal ratio thereto.
[0018] The modulation coils are preferably designed and configured for generating modulation frequencies which differ by more than their line width and by more than the line width of the high-frequency signal of the resonators. For example, the modulation frequencies of the plurality of modulation coils differ by more than 5% in each case, preferably by more than 20%.
[0019] In an advantageous embodiment, the modulation coils of the modulation device are arranged in the form of a one-dimensional, in particular linear, array. The one-dimensional array extends in particular transversely to a transport direction of the data carrier to be tested and enables a two-dimensional scan of a data carrier being moved in the transport direction.
[0020] An arrangement of multiple modulation coils one after another in the transport direction is also possible, in particular in combination with a spatially inhomogeneous polarization field, and can be used to generate a spectral resolution. In a further advantageous embodiment, the modulation coils of the modulation device are arranged in the form of a two-dimensional array, for example on the grid points of a regular grid, for example in a rectangular, hexagonal or line-by-line offset arrangement, and enable a two-dimensional spatial resolution even on a stationary data carrier.
[0021] The arrangement of the modulation coils extends advantageously over the entire width of the data carrier to be tested, in particular a banknote, in order to enable a test for completeness.
[0022] According to an advantageous embodiment, the resonator device has a plurality of stripline resonators, and in particular it is preferably provided that the number of stripline resonators in the resonator device is equal to the number of modulation coils in the modulation device, or that the number of stripline resonators in the resonator device is an integer multiple of the number of modulation coils in the modulation device.
[0023] The stripline resonators of the resonator device advantageously have the same resonance frequency, for example with a frequency deviation of less than 1%, preferably of less than 0.1%. Preferably, the stripline resonators of the resonator device are even designed and configured for operation in the same spatial mode. Alternatively or additionally, it is also advantageously provided that the stripline resonators of the resonator device have the same geometric shape, for example a square, rectangular or annular shape. The stripline resonators are preferably designed and configured for operation at the same excitation frequency, for example with a frequency deviation of less than 1%, preferably of less than 0.1%.
[0024] The stripline resonators of the resonator device are preferably arranged in the same plane, advantageously on a common resonator carrier, in particular a common printed circuit board. This plane is conveniently perpendicular to the direction of the static magnetic flux. Since the flat stripline resonators generate a B.sub.1 field with field components primarily in the plane of the resonators, the generated field is then perpendicular to the polarization field B.sub.0, as required for the spin resonance excitation.
[0025] In an advantageous variant of the invention, the polarization device generates substantially the same static magnetic flux at the location of each of the stripline resonators. The maxi-mum deviation of the static magnetic flux at the location of different stripline resonators is advantageously less than 2%.
[0026] In another, also advantageous variant of the invention, the polarization device generates a spatially inhomogeneous static magnetic flux in the air gap, for example in order to achieve a spectral resolution in the measurement.
[0027] In advantageous embodiments, multiple stripline resonators, in particular the same number in each case, are located in the region of the modulation field of each modulation coil. In the case of similar, jointly interconnected stripline resonators, this allows an improvement in the signal-to-noise ratio. In particular, each modulation coil can be assigned an N×M array of stripline resonators, where N and M are natural numbers and at least one of the values of N and M is greater

than 1, wherein the stripline resonators of the N×M array are all fed from the same signal source and are electrically connected in parallel and/or in series.

[0028] In a further advantageous embodiment, multiple stripline resonators with different resonance frequencies, in particular the same number in each case, are located in the region of the modulation field of each modulation coil. In particular, the stripline resonators are excited with different excitation frequencies that match the resonance frequency in each case. In addition to the spatial resolution, a spectral resolution can thus be achieved.

[0029] In another likewise advantageous embodiment, the sensor element contains only one or a small number of stripline resonators with extended field distribution, each covering the region of multiple modulation coils. The extended field distribution advantageously contains multiple localized field maxima, for example due to the operation of the single resonators in a higher spatial mode.

[0030] In advantageous embodiments, it is provided that the modulated magnetic field in the measuring range of each stripline resonator in has substantially only one single modulation frequency f.sub.Mod, i. However, the same modulation frequency f.sub.Mod,i can exist in the measuring range of several stripline resonators i.sub.1, . . . , i.sub.μ, . . . , i.sub.n with n≥1. In addition, further stripline resonators j.sub.1, . . . , j.sub.n are provided, in the measuring range of which the modulated magnetic field has a different modulation frequency f.sub.Mod,j. The crosstalk of this modulation field B.sub.Mod,j with the observed, for example adjacent, stripline resonator i.sub.μ is described by a contamination factor

[00001] $_{i\ j} = \frac{\int B_{\mathrm{Mod},j}\,\mathrm{dV}_{i_u}}{\int B_{\mathrm{Mod},i}\,\mathrm{dV}_{i_u}}$

wherein the integrals extend over the volume V.sub.iμ, which covers the sensitive region of the resonator i.sub.μ. The sum S.sub.iμ of the contamination factors at the location of the resonator in then describes the contribution of all modulation field components of other frequencies:

[00002]$S_{i_u} = .\mathrm{Math}._{j \neq i} \quad {}_{i\ j}$

[0031] This sum S.sub.iμ is advantageously less than 2%, in particular less than 0.5% for all stripline resonators i.sub.μ. This allows for a clean separation of the resonators based on their respectively assigned modulation frequencies.

[0032] The air gap advantageously has a height, i.e. a dimension in the z-direction, of less than 10 mm, preferably of less than 5 mm.

[0033] This allows a particularly strong polarization field, i.e. a strong static magnetic flux, to be generated in the air gap.

[0034] The invention also includes a test device for testing a flat-surface data carrier, in particular a banknote, having a spin resonance feature having a sensor element of the type described above and exactly one signal source from which all the stripline resonators of the resonator device are fed.

[0035] Advantageously, the test device also includes a transport device which inserts the flat-surface data carriers to be tested along a transport path into the air gap of the magnetic core or passes them through the air gap of the magnetic core, wherein the modulation coils of the modulation device are arranged in the form of a one-dimensional array which extends transversely to the direction of the transport path. The transport device is preferably configured for high-speed transport, for example between 1 m/s and 12 m/s, of the flat-surface data carriers.

[0036] The invention also includes a method for testing a flat-surface data carrier, in particular a banknote, having a spin resonance feature by means of a sensor element of the described type or a test device of the described type, wherein in the method [0037] a flat-surface data carrier to be tested is inserted into the air gap of the magnetic core of the aforementioned sensor element, [0038] the polarization device is used to generate a static magnetic flux in the air gap and the modulation device is used to generate a time-varying magnetic modulation field in the air gap, so that the modulated magnetic field generated by the modulation device together with the polarization device has different modulation frequencies at different locations within the air gap, and [0039] the

resonator is used to excite the spin resonance feature of the data carrier to be tested.
[0040] Advantageously, a response signal of the spin resonance feature generated by the excitation is also recorded with the resonator device and output to a detector. The excitation of the spin resonance feature and/or the recording of the response signal of the spin resonance feature can be carried out in a continuous wave (CW) method, in a pulsed method, or in a rapid scan method.
[0041] Further exemplary embodiments as well as advantages of the invention are explained below by reference to the figures, in the representation of which a true-to-scale and proportional reproduction has been omitted in order to increase the clarity.

Description

[0042] In the drawing:
[0043] FIG. **1** schematically shows a test device of a banknote processing system for the measurement of spin resonances of a banknote test specimen,
[0044] FIG. **2** schematically shows the design of the resonator device and the modulation device of a sensor element according to the invention in a first exemplary embodiment,
[0045] FIG. **3** shows the spectral power distribution of the RF signal reflected from the sensor element according to FIG. **2** for different modulation frequencies, and
[0046] FIG. **4** schematically shows the circuit wiring used in the described simulation of a sensor element with an array of two modulation coils and two square λ/2 stripline resonators.
[0047] The invention will now be explained using the example of the authenticity testing of banknotes. FIG. **1** schematically shows a test device **20** of a banknote processing system for the measurement of spin resonances of a banknote test specimen **10**.
[0048] The banknote test specimen **10** contains a spin resonance feature **12** to be tested, the characteristic properties of which are used to prove the authenticity of the banknote. For the authenticity test, the banknote test specimen **10** is guided along a transport path **14** through a sensor element **30** according to the invention of the test device **20**. For the detection of spin resonance signatures of the spin resonance feature **12**, the sensor element **30** generates three different magnetic fields in the measuring range.
[0049] Firstly, a static magnetic flux is generated parallel to the z-axis in the measuring range by a polarization device **34**. Secondly, a modulation device **36** generates a time-varying magnetic modulation field in the air gap, which also runs parallel to the z-axis and has modulation frequencies f.sub.Mod in the range between 1 kHz to 1 MHz. While conventional modulation devices usually generate only a single modulation frequency, the modulation device **36** according to the invention contains multiple modulation coils in the manner described in more detail below, which are designed and configured for generating different modulation frequencies.
[0050] Finally, a resonator device **32** generates an excitation field, which induces the energy transitions between the spin energy levels in the spin resonance feature **12**. The excitation field typically has frequencies above 1 GHz and is polarized perpendicular to the z direction.
[0051] The frequency of the excitation field is tuned to the Larmor frequency of the spin resonance feature **12** to be detected, in order to measure its spin resonance signature and to allow it to be used for the authenticity test. For this purpose, the test device **20** contains a signal source **22**, the excitation frequency few of which corresponds to the expected Larmor frequency of the spin resonance feature **12**. The excitation signal of the signal source **22** is supplied via a duplexer **24** to the resonator device **32**, where it generates a magnetic alternating field with frequency f.sub.MW.
[0052] In the present invention, the resonator device **32** comprises one or more stripline resonators for generating the excitation field in the manner described in more detail below. A stripline resonator is a conductive structure with a characteristic length **1**, which is applied to a substrate, such as a printed circuit board or a piece of ceramic. If the wavelength λ of the coupled-in high-

frequency signal on the printed circuit board matches the dimensioning of the conductor structure, a stationary wave can form, and the resonator is then in resonance at the frequency associated with λ. In particular, stripline resonators are characterized in that their sensitive region is very easily accessible and that they have a very high filling factor for flat samples, such as those formed by the banknotes to be tested.

[0053] In addition to the said elements, the test device **20** includes a detector diode **26** for measuring the high-frequency power reflected by the resonator device **32** and an evaluation unit **28** for evaluating and optionally displaying the measurement result. If the spin resonance feature **12** is in resonance at a coupled-in frequency f.sub.MW, the resonator quality changes, and with it the power reflected by the stripline resonators. Due to the modulation of the static polarization field by the modulation device **36**, the exact value of the Larmor frequency of the sample oscillates, so that the obtained measurement signal is amplitude-modulated with the modulation frequency.

[0054] FIG. **2** schematically shows the design of the resonator device **32** and the modulation device **36** of a sensor element **30** according to the invention according to a first exemplary embodiment of the invention. In this exemplary embodiment, the sensor element **30** contains a resonator device **32** consisting of an array of jointly connected stripline resonators **32-1**, **32-2** in combination with a modulation device **36** consisting of an array of modulation coils **36-1**, **36-2**.

[0055] In the figure, to illustrate the basic principle, an embodiment with only two resonators **32-1**, **32-2** and only two modulation coils **36-1**, **36-2** is shown, but it is understood that a larger number of resonators **32-***j*, with j=1, . . . , m, and of modulation coils **36-***i*, with i=1, . . . , n, with natural numbers n and m (m=number of stripline resonators, n=number of modulation coils) can be provided. While in the design of FIG. **2** the case n=m=2 is shown, in general m≥1 and n≥2 applies, where n can be equal to m, but also unequal to m.

[0056] The modulation coils **36-***i* of the modulation-coil array **36** in the exemplary embodiment generate for each resonator of the resonator device **32** a local modulation field with its own modulation frequency f.sub.Mod,i. This results in specific modulation peaks **44** and **46** in the spectral power distribution **40** of the reflected microwave signal for each resonator **32-***j*, as illustrated in FIG. **3**. These modulation peaks **44**, **46** can be separated from each other digitally, i.e. in a space-saving manner, by frequency in the lock-in method.

[0057] Specifically, in the exemplary embodiment of FIGS. **2** and **3**, due to the common circuit wiring **38** the resonators **32-1**, **32-2** receive an excitation signal of the same frequency f.sub.MW from the signal source **22**. In the modulation device **36**, the first modulation coil **36-1** generates a modulation field with the frequency f.sub.Mod,1, and the modulation coil **36-2** generates a modulation field with a different frequency f.sub.Mod,2≠f.sub.Mod,1.

[0058] The two modulation frequencies are chosen in such a way that they are not in a simple integer ratio and their difference is chosen so large that the modulation frequencies f.sub.Mod,1, f.sub.Mod,2 differ by more than their line width and by more than the line width of the high-frequency signal of the resonators **32-1**, **32-2**. In particular, the modulation frequencies even differ by more than twice the largest of the given line widths.

[0059] With reference to the power spectrum **40** of the reflected high-frequency signal shown in FIG. **3**, the carrier frequency f.sub.MW (reference sign **42**) is flanked by first modulation peaks **44**, which are offset by ±f.sub.Mod,1 relative to the carrier frequency **42**, and is flanked by two modulation peaks **46**, which are offset by ±f.sub.Mod,2 relative to the carrier frequency **42**. The modulation peaks **44**, **46** can be easily separated from each other by frequency using the lock-in method.

[0060] Within the scope of the invention, resonator devices **32** and modulation devices **36** with a fairly large number of stripline resonators or modulation coils are possible, as long as it is ensured that the various frequency components in the power spectrum can still be cleanly separated.

[0061] The exemplary embodiment of FIG. **2** shows an advantageous embodiment, in which the number m of the resonators **32-***j* is equal to the number n of the modulation coils **36-***i* and in which the resonators **32**-*j* and modulation coils **36**-*i* are assigned to each other in pairs. The modulation coils **36**-*i* are arranged in this exemplary embodiment in the form of a one-dimensional array **36**, which extends transversely to the transport direction **14** of the banknote **10**. In conjunction with the movement of the banknote test specimen, a two-dimensional scan of the transported banknote test specimen **10** is thereby enabled.

[0062] In addition to the parallel connection shown in FIG. **2**, a series connection or a combination of series and parallel connection are also possible for the resonators **32**-*i*. In another embodiment, a single resonator with extended field distribution, in particular with multiple localized field maxima, can cover the range of multiple modulation coils by operating in a higher spatial mode.

[0063] For the correct functionality of the modulation coil array, it is advantageous if the modulation coils "cross-modulate", i.e. contaminate the modulation channels of the adjacent modulation coils, as little as possible. Such contamination can occur when, for example, the modulation coil **36-1** of FIG. **2** generates a modulation field B.sub.Mod,1, which still has a substantial amplitude in the sensitive region of the resonator **32-2**, or conversely, the modulation coil **36-2** generates a modulation field B.sub.Mod,2 which still has a substantial amplitude in the sensitive region of the resonator **32-1**.

[0064] If there is an equal number k=n=m of resonators and modulation coils, a contamination factor χ.sub.ij can be defined by

[00003] $_{ij} = \frac{\int B_{\mathrm{Mod},j}\, \mathrm{d}V_i}{\int B_{\mathrm{Mod},i}\, \mathrm{d}V_i}$ with $j = 1$, .Math. , k, and $i = 1$, .Math. , k.

[0065] The contamination factor χ.sub.ij indicates the size of the parasitic signal components of other modulation frequencies, which are generated by a cross-modulation in the resonator **32**-*i*. The volume V.sub.i to be used for the integration describes the sensitive region of the resonator **32**-*i*.

[0066] For i=j, the variables χ.sub.ii are also formally defined and by design equal to 1, but they do not describe contamination, rather the desired signal generated by the modulation coil **36**-*i* in the associated resonator **32**-*i*, and are therefore disregarded in the summing operation.

[0067] For the functionality of the modulation coil array **36** it is now advantageous that for each modulation channel **36**-*i* the sum over all contamination factors, that is

[00004] .Math.$_{j}^{j \neq i}$ $_{ij}$

is less than 2%. The sum for each modulation channel is particularly advantageously even less than 0.5%.

[0068] An analogous definition of the contamination factors and the sum of the contamination factors can be used if the number m of resonators is not equal to the number n of modulation coils.

[0069] In order to demonstrate the superior performance of sensor elements according to the invention, the behavior of a sensor element according to FIG. **2** was simulated with an array of two modulation coils **36-1**, **36-2** and two square λ/2 stripline resonators **32-1**, **32-2**.

[0070] The circuit **50** underlying the simulation is shown schematically in FIG. **4**. It contains a digital part **52** and an analog part **54**.

[0071] The two λ/2 stripline resonators **32-1**, **32-2** are mounted on a printed circuit board with a thickness of 1.5 mm, the dielectric constant of which is 3.66. The resonators **32-1**, **32-2** are a distance of 15 mm apart, the edge length of the resonators is 7.1 mm in each case, corresponding to a resonance frequency of 9.8 GHz.

[0072] The impedance of each resonator **32-1**, **32-2** is transformed up to 100Ω by means of a 24 impedance transformer. A total impedance of 50Ω is obtained by connecting the two basic elements in parallel. The resonator array **32** formed from the two resonators is fed via a circulator **56** from the amplified output signal of a signal source **22**. The signal source **22** is operated at 9.8 GHz in continuous wave (CW) mode.

[0073] A planar modulation coil **36-1** or **36-2** is positioned opposite each resonator **32-1** and **32-2** at a distance of 2 mm. The modulation coils **36-1**, **36-2** are spiral-shaped, have 15 turns and a diameter of 5 mm. The modulation coil **36-1** of the first resonator **32-1** is operated with a frequency f.sub.Mod,1=20 kHz, the modulation coil **36-2** of the second resonator **32-2** with a frequency f.sub.Mod,2=30 kHz. The respective modulation signals are digitally generated in an FPGA, then subjected to a D/A conversion and amplified such that the same current flows through both modulation coils **36-1**, **36-2**.

[0074] The resonator array **32** and the modulation coil array **36** were placed in the air gap of a magnetic circuit and loaded with a paper sample provided with a spin resonance feature. The Larmor frequency of the spin resonance feature with the present polarization field corresponds exactly to the 9.8 GHz excitation frequency.

[0075] In the next step, the signal reflected from the resonator array **32** was amplified with a low-noise receiver amplifier **58** and downmixed with the 9.8 GHz excitation signal (reference sign **60**). The phase shifter and the filter banks are not shown in FIG. **4** for the sake of clarity. After downmixing, the signal is digitized and fed to the FPGA.

[0076] In the FPGA, the signal is split into two channels. Both channels are bandpass-filtered (reference sign **62**), with the first channel having a center frequency of 20 kHz and the second channel having a center frequency of 30 kHz. Both filters have a bandwidth of 5 kHz. The first channel is then demodulated with the 20 kHz modulation signal, the second channel with the 30 kHz modulation signal, and the demodulated output signals are fed to an evaluation unit **64-1** for channel **1** and an evaluation unit **64-2** for channel **2**. The demodulation involves a quadrature-amplitude modulation. The phase shifters used are again not shown in the figure. The associated filter banks, which are also not shown in the figure, have a bandwidth of 2.5 kHz.

[0077] Finally, the polarization field of the magnetic circuit was traversed with a field sweep and the output signals of the two channels were recorded by the evaluation units **64-1**, **64-2**. Both channels show the spectrum of the spin resonance feature used for the doping and correspond to different measuring points on the banknote.

TABLE-US-00001 List of reference signs 10 banknote test specimen 12 spin resonance feature 14 transport path 20 test device 22 signal source 24 duplexer 26 detector diode 28 evaluation unit 30 sensor element 32 resonator device 32-1, 32-2, 32-j resonators 34 polarization device 36 modulation device 36-1, 36-2, 36-i modulation coils 40 spectral power distribution 42 carrier frequency 44, 46 modulation peaks 50 circuit 52 digital circuit part 54 analog circuit part 56 circulator 58 reception amplifier 60 downward mixing 62 bandpass filtering 64-1, 64-2 evaluation units

Claims

1.-**19**. (canceled)

20. A sensor element for checking a flat-surface data carrier having a spin resonance feature, with a magnetic core with an air gap, into which the flat-surface data carrier can be inserted for testing, a polarization device for generating a static magnetic flux in the air gap, a resonator device for exciting the spin resonance feature of the data carrier to be tested in the air gap, having at least one stripline resonator fed by a signal source, and a modulation device for generating a time-varying magnetic modulation field in the air gap parallel to the static magnetic field, wherein the modulation device comprises a plurality of modulation coils, which are designed and configured for generating different modulation frequencies, so that the modulated magnetic field generated by the modulation device together with the polarization device has different modulation frequencies at different locations within the air gap.

21. The sensor element according to claim 20, wherein the modulation coils of the modulation device are arranged offset next to one another.

22. The sensor element according to claim 20, wherein at least one modulation coils of the modulation device are formed by planar coils, which have one or more turns around an axial direction of the air gap in a plane.

23. The sensor element according to claim 20, wherein the modulation coils are arranged in the same plane on a common coil carrier.

24. The sensor element according to claim 20, wherein the modulation coils are designed and configured for generating modulation frequencies which are not in a simple integer ratio to each other.

25. The sensor element according to claim 20, wherein the modulation coils are designed and configured for generating modulation frequencies which differ by more than their line width and by more than the line width of the high-frequency signal of the resonators.

26. The sensor element according to claim 20, wherein the modulation coils of the modulation device are arranged in the form of a one-dimensional array.

27. The sensor element according to claim 20, wherein the arrangement of the modulation coils extends over the entire width of the data carrier to be tested.

28. The sensor element according to claim 20, wherein the resonator device has a plurality of stripline resonators, with the number of stripline resonators in the resonator device being equal to the number of modulation coils in the modulation device, or that the number of stripline resonators in the resonator device is an integer multiple of the number of modulation coils in the modulation device.

29. The sensor element according to claim 20, wherein the stripline resonators of the resonator device have the same resonance frequency, with the stripline resonators of the resonator device designed and configured for operation in the same spatial mode, and that the stripline resonators of the resonator device have the same geometric shape.

30. The sensor element according to claim 20, wherein the stripline resonators of the resonator device are arranged in the same plane, advantageously on a common resonator carrier.

31. The sensor element according to claim 30, wherein said plane is perpendicular to the direction of the static magnetic flux.

32. The sensor element according to claim 20, wherein the polarization device generates substantially the same static magnetic flux at the location of each of the stripline resonators.

33. The sensor element according to claim 20, wherein the polarization device generates a spatially inhomogeneous static magnetic flux in the air gap.

34. The sensor element according to claim 20, wherein the modulated magnetic field has substantially only one single modulation frequency in the measuring range of each stripline resonator.

35. The sensor element according to claim 20, wherein the air gap has a height of less than 10 mm.

36. A test device for testing a flat-surface data carrier having a spin resonance feature, having a sensor element according to claim 20, and exactly one signal source from which all stripline resonators of the resonator device are fed.

37. The test device according to claim 36, having a transport device which inserts the flat-surface data carriers to be tested along a transport path into the air gap of the magnetic core or passes them through the air gap of the magnetic core, wherein the modulation coils of the modulation device are arranged in the form of a one-dimensional array which extends transversely to the direction of the transport path.

38. A method for testing a flat-surface data carrier having a spin resonance feature, by means of a sensor element or by means of a test device according to claim 36, wherein in the method a flat-surface data carrier to be tested is inserted into the air gap of the magnetic core of the aforementioned sensor element, the polarization device is used to generate a static magnetic flux in the air gap and the modulation device is used to generate a time-varying magnetic modulation field in the air gap, so that the modulated magnetic field generated by the modulation device together with the polarization device has different modulation frequencies at different locations within the air gap, and the resonator device is used to excite the spin resonance feature of the data carrier to be tested.